Chapter 1 · Relations, keys and constraints

A college keeps a register with student number, name and department. Two students can share a name. A student number identifies one student. A separate enrolment list records which courses each student takes.

Student number	Name	Department
11	Asha	Physics
12	Ravi	Mathematics
13	Asha	Chemistry

Vocabulary. A relation is a set of tuples, presented as rows. An attribute is a named column. Its domain is the set of permitted values. A schema specifies attributes and constraints; an instance is the current collection of rows. Degree counts attributes; cardinality counts tuples. SQL tables may contain duplicate rows unless constraints prevent them.

A superkey is an attribute set that uniquely identifies a tuple in every permitted instance. A candidate key is a minimal superkey. Minimal means that no proper subset is a superkey; it does not mean the shortest key. A primary key is a selected candidate key. A prime attribute belongs to at least one candidate key. A foreign key references a permitted unique identifier in another table; referencing it does not make the foreign key unique.

R01 Identify keys and distinguish schema from instance

Use when: a question asks which attributes are keys.

1. List the stated uniqueness rules and dependencies.
2. Check whether a proposed set determines all attributes. Use closure if dependencies are supplied.
3. Remove attributes to test minimality.
4. Separate candidate keys from their larger superkeys.

Seeing three different names in a table does not establish that Name is a schema key. Conversely, two identical names disprove uniqueness for that instance. In SQL, a primary key is unique and non-null. Nullable UNIQUE constraints require care: their treatment of NULL is dialect-dependent.

Worked example W01. In the college register above, StudentID is the declared identifier. {StudentID} and {StudentID, Name} both identify rows. Only the first is minimal. Name is not a key: students 11 and 13 share it. If the current three department names differ, that coincidence still does not make Department a schema key. A relation with three attributes and three rows has degree 3 and cardinality 3. Adding a fourth student changes the cardinality, not the degree.

Check before moving on. Can a candidate key contain two columns? Yes. Can one table have two different candidate keys? Yes. Primary chooses one of them.

R02 Foreign key operations

Use when: insertion, deletion or updating may violate a reference.

1. Draw an arrow from the child reference to the parent identifier.
2. For child insertion/update, check that each required non-null reference has a matching parent.

3. For parent deletion/key update, check whether children still reference the old value.
4. Apply stated CASCADE, SET NULL or other actions. Without such an action, do not silently repair a broken reference.
5. Check other constraints independently.

Adding a parent or deleting a child cannot create a dangling reference under that foreign key alone. The operation may still violate another constraint. A foreign key is not automatically NOT NULL.

Worked example W02. Let Department contain IDs {10,20}. Student has rows (11,10), (12,20), (13,NULL), and Student.DeptID is a nullable foreign key. Inserting (14,30) creates a missing reference and fails. Inserting Department 30 first makes that reference valid. Deleting Student 11 is safe for this foreign key. Deleting Department 20 leaves Student 12 without a parent and fails unless a specified referential action repairs it. Under SET NULL it changes Student 12's reference to NULL, provided that column permits NULL. Inserting a parent is safe for this reference but can still violate the parent's primary key.

Check before moving on. Name the child and parent before deciding whether an update is safe. Do not assume a foreign key is unique or non-null.

R03 Possible table sizes and duplicate identifiers

Use when: a question asks for minimum/maximum rows or join sizes under keys.

Translate each key into an upper bound on matching rows. A child can reference a parent many times unless uniqueness prevents it. Total participation supplies lower bounds; keys often supply upper bounds. Construct an instance achieving a proposed bound before calling it attainable.

Worked example W03. Four purchase rows reference CustomerID 1,1,2,NULL. Customer has IDs 1 and 2, with CustomerID a primary key. The inner join returns three rows: each non-null purchase has exactly one parent, while NULL has no equality match. The left join returns four rows, including the unmatched purchase. Joining in the other direction does not give one row per customer: customer 1 appears twice. If all four purchase references were non-null valid foreign keys, the inner join would have exactly four rows. Without the foreign key, the upper bound would still be four, but the lower bound could be zero.

Check before moving on. State which side is unique and whether every reference must match. Then count matches, not just source rows.

Previous-year questions, worked

Read the original question and options in the linked official paper. These are original explanations, not replacement question statements. PDF page numbers count from the first page of the downloaded file.

DA-2025-Q56 · Candidate keys and NULL

Official question, PDF page 30. Published key: **B; C**.

Customer's primary key is (name,email). Its aadhaar column is only a foreign key, so repeated references are allowed: A is false. Phone has no NOT NULL restriction: B is true. Person.aadhaar is the primary key and therefore a candidate key: C is true. A primary key cannot contain NULL: D is false. Do not transfer a uniqueness constraint from the referenced table to the referencing column.

DA-2026-Q26 · Safe foreign-key operations

Official question, PDF page 18. Published key: **A; B**.

Inserting into the referenced parent cannot remove an existing target. Deleting from the referencing child cannot create a dangling reference. These are A and B. Inserting a child can reference a missing parent. Deleting a parent can leave an existing child without its target. Those operations are not always safe. The conclusion concerns the named foreign key alone, not every possible constraint.

CS-2024-set1-Q22 · First normal form and keys

Official question, PDF page 15. Published key: **A; C**.

Under the exam's atomic-value convention, 1NF permits a multi-attribute key, so A is true. It does not prohibit foreign keys, so B is false. A composite attribute must be represented through its atomic components rather than one structured cell, so C is the intended true statement. Several candidate keys are permitted, so D is false. Atomicity concerns a column's chosen domain; it is different from a key using several columns.

CS-2023-Q16 · Degree versus cardinality

Official question, PDF page 13. Published key: **A**.

Count the attributes in the schema. That count is the degree. Count tuples in an instance for its cardinality. The option identifying the attribute count is A. Neither the number of possible values nor the number of currently stored rows defines degree.

Chapter practice

Original questions. Write the answer and the intermediate reasoning. Each lettered part belongs to the same problem. The separate solutions guide contains hints, steps and a final check.

P01 · Short answer

Student(ID, Name, Email) has declared candidate keys {ID} and {Email}. The current instance has 4 rows and distinct names. (a) List all candidate keys and all superkeys. Give its degree and cardinality. (b) Is {Name} a schema key? If a fifth row is added, which count changes?

P02 · MSQ plus variation

Parent has primary keys {1,2}. Child has (ID,ParentID) rows (10,1),(11,2),(12,NULL), with a nullable foreign key and no cascade action. Which operations are guaranteed safe for this foreign key alone? A: add a fresh parent 3. B: delete child 10. C: add child (13,9). D: delete parent 2. (b) Repeat D when ON DELETE SET NULL is specified and the child column remains nullable.

P03 · NAT plus variation

Customer has unique IDs {1,2,3}. Purchase has references {1,1,2,NULL}; its non-null references form a valid foreign key. (a) Count the inner join rows and Purchase LEFT JOIN Customer rows. (b) Add a purchase with reference 3. Recompute both counts. Does the inner join list each customer once?

Before the exam

Be able to distinguish schema from instance, minimal from minimum, and candidate from primary key. For a reference operation, identify what could become missing. For a row-count bound, state the uniqueness and participation assumptions that make the bound attainable.